

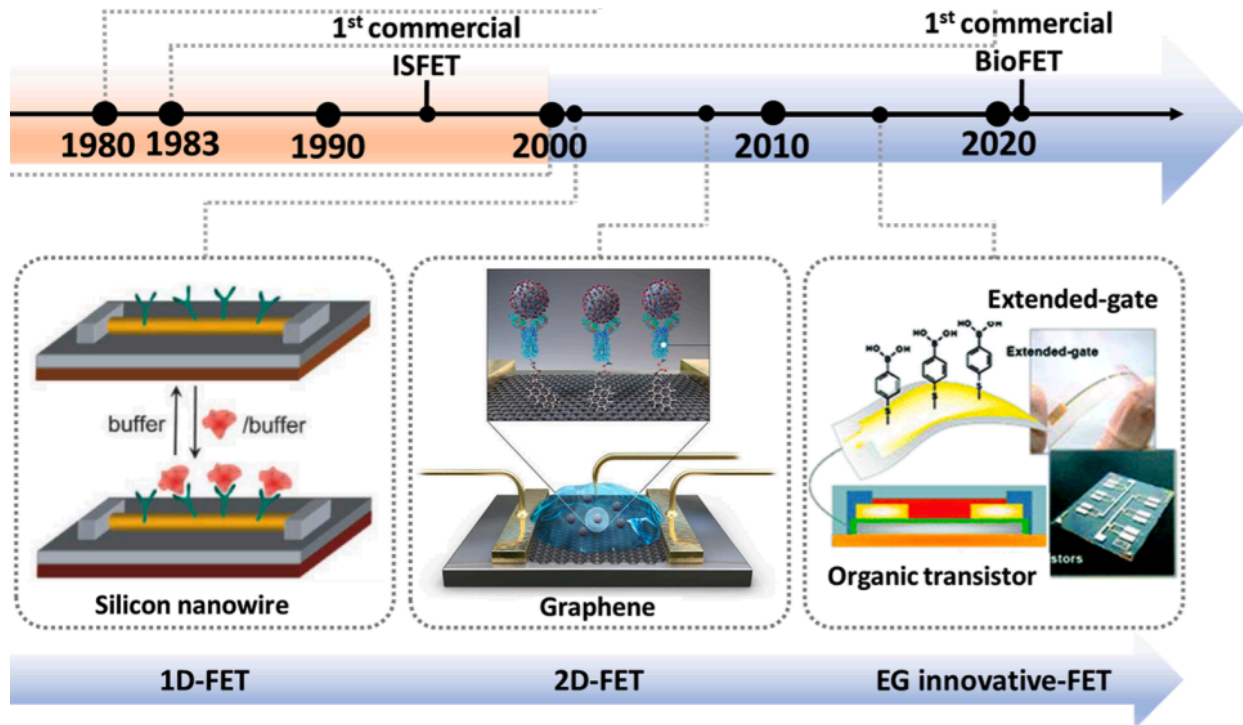
Sundew Research

Electrical interface:

Extended Gate Field-Effect Transistor (EG-FET)

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- To serve PoC goals and become a part of the diagnostic toolkit in the future, such sensors must satisfy the set of requirements including high sensitivity, accuracy, rapid detection, and specificity toward the analyte of interest while remaining small-sized and affordable.
- Chemical sensors (or chemosensors) are devices that transform analyte information ranging from concentration to total composition analysis into analytically useful signals
- Biosensors, which used to be considered as a subgroup of chemosensors [5], have rapidly proliferated
- We want → FET-based biosensors



- EG-FET also brings some disadvantages in comparison with the ISFET, such as the inevitably larger device footprint and decrease in sensitivity due to the partial loss of the signal caused by transmission through the electrical connection line. However, such solution still offers a simple and low-cost architecture that may be highly relevant for PoC applications
- Remarkable results in EG-FET-based biosensor development have been demonstrated by several groups in the last few years illustrating excellent sensitivity, high accuracy, and promising diagnostic prediction capabilities

- EG-FET still suffer from several disadvantages limiting their cost-effectiveness and use in the PoC setting. Namely, EG-FET-based biosensing systems continue to rely on complex micro-/nano-fabrication of the extended gate (EG) and FET transducer to achieve high sensitivities and low detection limits
- Operation of these systems often requires broad voltage sweeping ranges (>10 V) and measurement at low current levels (in the nA range) leading to the demand for interfacing with bulky high-performance measurement instruments (semiconductor analyzers or source measure units).
- Namely, multiple strategies are under development to: (a) reach high sensitivity levels and low detection limits, while minimizing the complex processing for nanostructuring; (b) create stable standalone EG-FET-based biosensing platforms for PoC applications, while avoiding cumbersome, expensive, and specialized high-performance instruments; (c) achieve miniaturization and adaptation of the reference electrode to the measurement approaches, enabling further exploration of multiplexed sensing capabilities.
- Spatial separation between the sensing element and the transducer offers a two-sided advantage: (1) it improves the robustness of the measuring process by protecting the FET transducer from extraneous effects and (2) makes the measurements more affordable by allowing for a cost-effective disposable sensor component.
- physical separation of the EG electrode comes with certain disadvantages such as larger device footprint, reduced sensitivity caused by the losses during signal transfer, and increased noise or interference coupling
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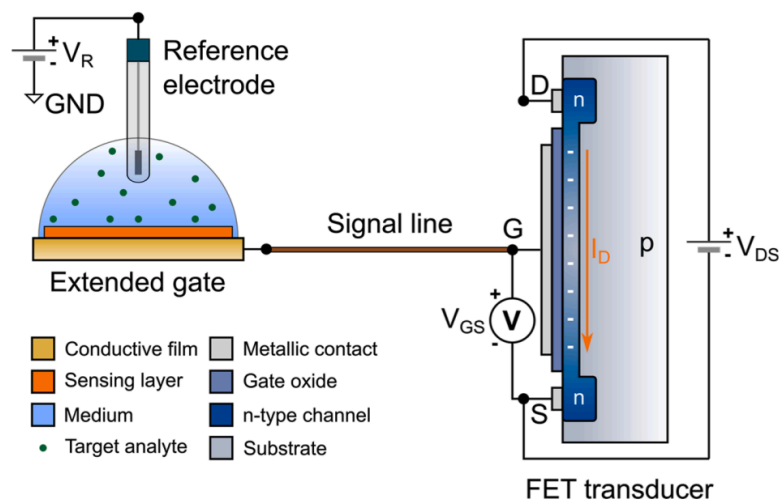


Fig. 3. Illustration of the EG-FET sensing concept using the example of n-type enhancement mode MOSFET transducer. Extended gate (EG) electrode equipped with a sensing layer is electrically connected to the gate terminal of the FET transducer. Biasing of the reference electrode and FET transducer enables potentiometric monitoring of surface potential shifts due to the interaction between the sensing layer and target analyte in electrolyte medium. (a) Abbreviations: GND - ground, D - drain, G - gate, S - source. (b) Symbols: V_R - reference electrode voltage, V_{GS} - gate-to-source voltage, V_{DS} - drain-to-source voltage, I_D - drain current.

→Reference electrode acts as a ground, staying constant and provides a known potential. Not affected by the biomolecules and sets the voltage baseline for the system

→Conductive Film/Sensing Layer is what changes when biomolecules interact with the aptamer.

→The change in voltage caused by this interaction is then measured by the FET's gate terminal (G). This change directly impacts the current flow through the MOSFET, and that change is read as a biosensor signal.

- Eq. (1) holds if the surface potential shift is mainly determined by the amount of charge deposited on the surface of the sensing layer and capacitance of the electrical double layer
 - However, in some FET sensor configurations, the effects of coupling with the FET device capacitance cannot be neglected, and differences in capacitive coupling between multiple gates can even be exploited to improve the sensing response
- A brilliant approach, acting as an alternative to the encapsulation of recognition elements within polymer matrices, is the use of molecularly imprinted polymers (MIPs) as recognition layers. The target analytes (template molecules) are included in a mixture of functional monomers for polymerization and later removed, leaving cavities that are specific for the same molecules of interest. Depending on the composition of MIP, the monomers assemble around template molecules by means of weak and medium interactions

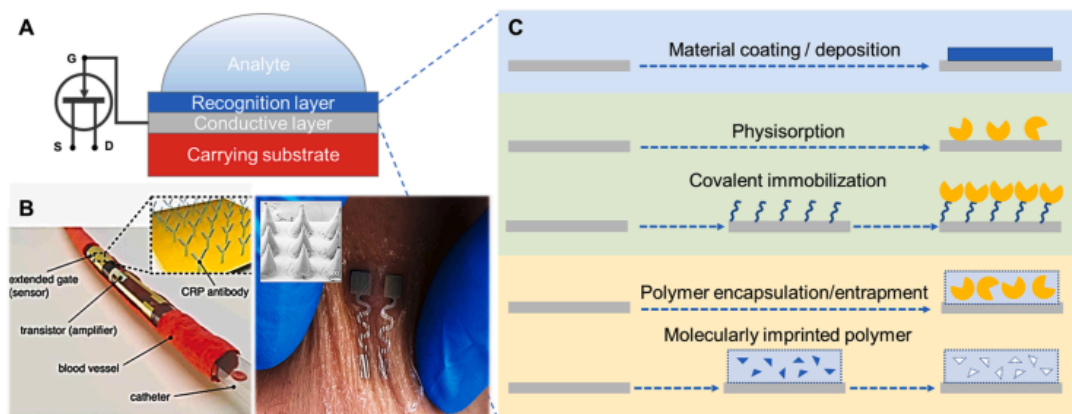


Fig. 4. Extended gate (EG) sensing component of the EG-FET sensor. (A) Building blocks of the EG sensing element. (B) Examples of flexibility in designing sensing components with EG-FET configuration; Left side: A smart surgical catheter with antibodies functionalized on a Au layer for C-reactive protein determination in blood [35]; Right side: Wearable microneedle with sodium ionophore encapsulated in poly(vinyl chloride) for continuous sodium monitoring in interstitial fluids [34]. (C) Strategies for producing recognition layers to selectively capture specific analytes. All illustrations are adapted with permission.

- Lately, the use of MIPs has been demonstrated in EG-FET sensors with very promising results in the sensing of small proteins and peptide molecules
- In EG-FET sensors, FETs are employed as transducers and they convert analyte interactions into detectable electrical signals.
- The semiconductor forms a channel between the source and the drain which has excess of one charge carrier type (holes in p-type and electrons in n-type). The presence of a high- κ dielectric as an insulator between the channel and the metal gate in the FET enables the gate to modulate charge carrier concentration within the channel, while simultaneously preventing the carriers from traversing through the gate, thus minimizing the leakage current.

- Occasionally, a bottom gate (also known as a body or back gate), is utilized to drive the transistor. Driving of the FET using a bottom gate comes at the cost of higher power and stronger biasing required to operate the device. However, the combined use of both top and bottom gates, commonly referred to as a dual-gate configuration, can substantially enhance measurement sensitivity [56,57]. This amplification of the signal is achieved via capacitive coupling between the top and bottom gate dielectrics, and it can be regulated by adjusting the ratio of top and bottom gate capacitance
- In contrast to an ISFET, the FET in EG-FET configuration is physically separated and protected from the electrolyte solution, as well as external factors such as light and temperature, when appropriately packaged.
- Conventional inorganic FETs, here referred to as commercial FETs, can also be used as transducers in an EG-FET configuration. They offer a straightforward, fast, and efficient solution that finds widespread use in ion and pH sensing, as well as in assays involving redox reactions (like, e.g., in enzyme-based FET sensors). This option is known for its high reproducibility and cost-effectiveness.
- It has been demonstrated that higher sensitivity can be achieved in EG-FET format by using nanostructured FETs
- Compared to organic materials, inorganic materials offer superior performance in terms of charge carrier mobility, higher switching rates, larger ION/IOFF, and better reliability. However, research on organic FETs (OFETs) is evolving rapidly and offers a wide range of materials and device configurations with performance sufficient for many sensing applications
- Moreover, unlike inorganic materials which require demanding fabrication facilities, such as high-tech lithography, dry and wet etching, and high-temperature treatment, OFETs can be fabricated using relatively simple and low-cost techniques for solution processing [68], such as spin coating and inkjet printing, which allows for more flexible and cost-effective manufacturing options.

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- "Feasibility Study of Extended-Gate-Type Silicon Nanowire Field-Effect Transistors for Neural Recording"
→ <https://www.mdpi.com/188674>
- "Design and Development of Ion-Sensitive Field-Effect Transistor and Extended-Gate Field-Effect Transistor Platforms for Chemical and Biological Sensors"
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